

Standard Operating Procedure

Concrete Floor Drilling – Utility Strike Prevention

Document No.	SOP-LV-002
Related Documents	SOP-LV-003, SOP-LV-006, IR-2026-07-08-001
Effective Date	July 13, 2026
Revision	Rev. 0 (reconstructed)
Jurisdiction	Ontario (O. Reg. 213/91 – Construction Projects; WSIB)
Applies To	All field technicians performing drilling or coring into concrete floors or slabs
Approved By	_____

NOTE ON THIS DOCUMENT:

This SOP was reconstructed after the original file could not be located. It follows the same topic and structure as the original but is newly drafted — review carefully before relying on it for training or compliance purposes.

1. Purpose

This procedure establishes the technical steps for safely drilling or coring into concrete floors and slabs, including non-destructive detection methods, drill site marking, and progressive drilling technique. It works together with SOP-LV-006 (Utility Strike Prevention), which covers customer notification and stop-work authority — this document covers the hands-on drilling procedure itself. It was issued following a July 8, 2026 incident at Elsie McGill Public School in which a technician struck an in-floor heating pipe while drilling (IR-2026-07-08-001).

2. Scope

This SOP applies to all drilling, coring, or anchoring work into concrete floors, slabs, or block walls performed by Company technicians and subcontractors across Ontario and Manitoba. For notification requirements and stop-work authority, see SOP-LV-006. For wall-mounted equipment anchoring into non-concrete walls, see SOP-LV-006 Section 6.

3. Responsibilities

Site Supervisor

- Confirms utility verification per SOP-LV-006 Section 4 is complete before drilling is scheduled.
- Provides or arranges access to non-destructive detection equipment where required.
- Reviews and retains documentation of utility verification for the site file.

Field Technician

- Does not begin drilling at any location until the steps in this procedure are complete for that specific location.
- Uses the progressive drilling technique in Section 6 for all concrete floor penetrations.
- Stops immediately and reports any resistance, moisture, or unexpected material encountered while drilling.

4. Pre-Drilling Verification

Before drilling begins at a location, confirm the following have been completed per SOP-LV-006:

- Customer/facility contact has been asked about concealed services (in-floor heating, plumbing, electrical, post-tension cables, fire suppression piping) and the response is documented.
- Available mechanical, electrical, or as-built drawings for the work area have been reviewed.
- Where drawings are unavailable or services cannot be ruled out, non-destructive detection per Section 5 has been performed.

Do not proceed to Section 6 until pre-drilling verification is complete and documented.

5. Non-Destructive Detection Methods

Ground-Penetrating Radar (GPR)

- Where available, use GPR to scan the drill area and identify rebar, post-tension cable, conduit, and piping before marking drill locations.
- Mark all detected objects directly on the floor surface using detection-appropriate marking (e.g., non-permanent marker or tape) before drilling.

Utility / Rebar Locator

- Use an electronic locator capable of detecting metallic pipe, conduit, and rebar within the scan area.
- Scan in a grid pattern across the full area to be drilled, not just the immediate drill point, to account for diagonal or offset runs.

Thermal Imaging

- Where in-floor heating is present or suspected, use a thermal imaging camera with the heating system running to visualize loop layout before selecting drill locations.
- Allow sufficient time for the heating system to establish a visible thermal pattern before scanning.

Where none of the above methods are available or conclusive, a small pilot hole per Section 6 with immediate withdrawal on resistance is the minimum acceptable verification step — this is a fallback, not a substitute for detection equipment where the situation warrants it (institutional sites, unknown floor construction, prior incident history at the site).

6. Drill Site Marking & Layout

- Mark all confirmed drill locations only after detection and drawing review is complete.
- Maintain clearance from any detected or marked service per the manufacturer/detection equipment's stated accuracy tolerance — do not drill directly adjacent to a marked line.
- Where multiple holes are required, mark and verify each location individually; do not assume clearance at one point extends to an adjacent point.

7. Progressive Drilling Procedure

For all concrete floor and slab penetrations, use a progressive (pilot-hole) technique rather than drilling directly to full depth:

- Drill a small-diameter pilot hole first, advancing slowly and monitoring for resistance, vibration change, moisture, or debris colour/texture change.
- Stop immediately and withdraw the bit if resistance increases unexpectedly, moisture appears, or drilling behaviour changes from what is expected for plain concrete.
- If the pilot hole completes cleanly, proceed to the full-diameter hole using the same location and monitoring the same way.
- Do not apply excessive downward force to push through resistance — stop and reassess instead.

8. Core Drilling Specific Precautions

- For wet coring, manage water discharge to avoid creating a slip hazard; use a vacuum shroud or containment ring where available.
- Secure the core rig base per manufacturer instructions before starting; do not operate a core drill without the base properly anchored or weighted.
- Apply the same pre-drilling verification and progressive technique as Section 7 — core drilling does not reduce the need for utility verification.
- Where a core drill encounters an obstruction consistent with rebar or embedded reinforcement, stop and reassess the location rather than forcing through.

9. If a Strike Occurs

- Stop work immediately at the location.
- Make the area safe (de-energize if electrical, shut off supply if plumbing, ventilate if applicable) without attempting repairs.
- Notify the Site Supervisor immediately.
- Do not disturb the site beyond what is needed for safety, pending assessment.
- Complete an incident report per company procedure and notify the facility/customer contact.

10. Documentation Requirements

For each drilling location or work area, the technician or supervisor must retain a record of: the utility verification method(s) used, drawings reviewed (if any), customer confirmation of concealed services, and any detection equipment readings. This record should be filed with the job documentation and referenced on the SOP-LV-003 jobsite safety checklist for that visit.

11. References

- Ontario Regulation 213/91 – Construction Projects (Occupational Health and Safety Act)
- WSIB – Workplace Safety and Insurance Board reporting requirements
- Canadian Electrical Code (CEC) / CSA standards, as applicable
- SOP-LV-003 – Ontario Jobsite Health & Safety Checklist
- SOP-LV-006 – Utility Strike Prevention – In-Floor Heating, Concealed Services & Wall-Mounted Equipment
- IR-2026-07-08-001 – Incident Report, Elsie McGill Public School (July 8, 2026)

12. Technician Sign-Off

By signing below, the technician confirms they have read, understood, and agree to follow this procedure before drilling or coring into any concrete floor or slab.

Printed Name	Signature	Date